Part 1: Stern-Volmer relation of quenching

$$\frac{F_0}{F} = 1 + K_{SV} \cdot [H]$$

This formula applies to both static and dynamic quenching. What differs is how K_{SV} depends on the free host concentration $[H]$ in each case.

For static quenching, expressed with the dissociation constant K_d and bound fraction θ respectively,

$$K_{SV} = K_a = \frac{1}{K_d}$$

$$\frac{F_0}{F} = 1 + \frac{[H]}{K_d} = \frac{1}{1-\theta}$$

For dynamic quenching,

$$K_{SV} = k_q \cdot \tau_0$$

$$\frac{F_0}{F} = 1 + k_q \cdot \tau_0 \cdot [H]$$

So, for pure dynamic quenching only, and for pure static quenching only, both SV should be linear ($\propto [H]$).

And the upward curve arises from a mixture of dynamic and static quenching:

$$\frac{F_0}{F} = \left(1 + k_q \cdot \tau_0 \cdot [H]\right) * (1 + K_a \cdot [H])$$

And this expanded quadratic function leads to a hyperbola curve (**upward curvature towards the y-axis**)

For the **upward curvature towards the x-axis**, the most likely reason is incomplete static quenching, with relative brightness of the dim complex (α)

The fluorescence of the system now consists of two components: unbound free dye and dim bound complex

$$F = F_0 * (1-\theta) + \alpha * F_0 * \theta$$

rewrite as:

$$\frac{F_0}{F} = \frac{1}{1 - \theta * (1-\alpha)}$$

The general formula concerning both dynamic quenching and dim species in static quenching is:

$$\frac{F_0}{F} = \left(1 + k_q \cdot \tau_0 \cdot [H]\right) * \frac{1}{1 - \theta * (1-\alpha)}$$

Part 2: Solving the bound fraction of host-dye complexation ($\boldsymbol{\theta}$)

For 1:1 host-dye complexation, the accurate bound fraction $\boldsymbol{\theta}$ is solved as:

$$\boldsymbol{\theta} = \frac{[\boldsymbol{H}]}{\boldsymbol{K_d} + [\boldsymbol{H}]} = \frac{([\boldsymbol{I_0}] + [\boldsymbol{H_0}] + \boldsymbol{K_d}) - \sqrt{([\boldsymbol{I_0}] + [\boldsymbol{H_0}] + \boldsymbol{K_d})^2 - \boldsymbol{4}[\boldsymbol{I_0}] * [\boldsymbol{H_0}]}}{\boldsymbol{2}[\boldsymbol{I_0}]} \approx \frac{[H_0]}{K_d + [H_0]}$$

Where $[H]$ is the free host concentration, when $[H_0]$ is added as the titrant.

More generally, for 1:n host-dye complexation, assume independent binding sites, and the accurate bound fraction $\boldsymbol{\theta}$ is solved as:

$$[I_0] = [I] + n * \theta * [H_0] \quad (mass\ balance)$$

$$\boldsymbol{\theta} = \frac{\boldsymbol{n}[\boldsymbol{H}]}{\boldsymbol{K_d} + \boldsymbol{n}[\boldsymbol{H}]} = \frac{([\boldsymbol{I_0}] + \boldsymbol{n}[\boldsymbol{H_0}] + \boldsymbol{K_d}) - \sqrt{([\boldsymbol{I_0}] + \boldsymbol{n}[\boldsymbol{H_0}] + \boldsymbol{K_d})^2 - \boldsymbol{4}[\boldsymbol{I_0}] * \boldsymbol{n}[\boldsymbol{H_0}]}}{\boldsymbol{2}[\boldsymbol{I_0}]} \approx \frac{n[H_0]}{K_d + n[H_0]}$$

Hill equation for non-cooperative binding:

$$\theta = \frac{(n[H_0])^h}{{K_d}^h + (n[H_0])^h}$$

Part 3: Build up the fitting equations

In the fitting page "Stern-Vomer" concerning both static and dynamic quenching, it is necessary to calculate the free host concentration responsible for dynamic quenching.

So, define $\boldsymbol{x}$ as the total dye concentration bound in the complex $\boldsymbol{HI_n}$

$$\boldsymbol{x} = \boldsymbol{\theta} * [\boldsymbol{I_0}] = \frac{([I_0] + n[H_0] + K_d) - \sqrt{([I_0] + n[H_0] + K_d)^2 - 4[I_0] * n[H_0]}}{2}$$

The free host concentration is written as:

$$[\boldsymbol{H}] = [\boldsymbol{H_0}] - \frac{\boldsymbol{x}}{\boldsymbol{n}}$$

Therefore, the final Stern-Volmer formula is:

$$\frac{\boldsymbol{F_0}}{\boldsymbol{F}} = \left(\boldsymbol{1} + \boldsymbol{k_q} \cdot \boldsymbol{\tau_0} \cdot [\boldsymbol{H}]\right) * \frac{\boldsymbol{1}}{\boldsymbol{1} - \boldsymbol{\theta} * (\boldsymbol{1} - \boldsymbol{\alpha})} = \frac{\boldsymbol{1} + \boldsymbol{k_q} \cdot \boldsymbol{\tau_0} \cdot \left([\boldsymbol{H_0}] - \frac{\boldsymbol{x}}{\boldsymbol{n}}\right)}{1 - \frac{\boldsymbol{x}}{[\boldsymbol{I_0}]} * (\boldsymbol{1} - \boldsymbol{\alpha})}$$

And the generalized fluorescence quenching formula is converted to

$$\boldsymbol{F} = \frac{\boldsymbol{F_0} * \left(1 - \frac{\boldsymbol{x}}{[\boldsymbol{I_0}]} * (\boldsymbol{1} - \boldsymbol{\alpha})\right)}{\boldsymbol{1} + \boldsymbol{k_q} \cdot \boldsymbol{\tau_0} \cdot \left([\boldsymbol{H_0}] - \frac{\boldsymbol{x}}{\boldsymbol{n}}\right)}$$